

Supplemental Material: Inverse Rendering for Modeling with Line Primitives

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1 Additional Capture Details

Table A1 summarizes the statistics of our *Fuzzy* dataset. Given camera poses estimated by COLMAP [Schonberger and Frahm 2016], we reconstruct a coarse surface using NeuS2 [Wang et al. 2023] to initialize line primitives around it. Although NeuS2 efficiently recovers a coarse object outline within approximately three minutes, it often produces noisy signed distance fields in low-texture black background regions, leading to cluttered surfaces extracted with the marching cubes algorithm [Lorensen and Cline 1987]. To mitigate this, we apply a simple manual preprocessing step that roughly aligns a bounding cube to the object and crops the reconstructed mesh by mesh intersection. This is performed once per scene and prevents noisy background regions from affecting the initial line distribution, particularly the line lengths initialized from nearest-neighbor distances between the seed points. We will additionally release these coarse proxy surfaces. Note that cube intersection may not fully remove noisy floating surface components. We allow the residual noise, as it can later be removed via our opacity optimization.

2 Additional Optimization Details

Initialization. After reconstructing the coarse surface, we uniformly sample seed points within a distance band of 0.1. We determine the band distance of 0.1 experimentally and use it for all datasets. Rather than concentrating samples tightly around the reconstructed surface, we intentionally use a conservatively wide band to avoid missing fine structures that surface reconstruction methods often fail to recover when applied to fuzzy geometry. To efficiently perform rejection sampling within this distance band, we first evaluate distances on a coarse voxel grid and then apply stratified sampling to generate candidates with a higher acceptance rate than ambient-space sampling. This strategy enables the sampling of 1M–2M points within a few seconds. We note that scenes in the *Shelly* dataset are normalized independently; although their normalized extents are similar, distances in the normalized coordinate system may not correspond directly to physical dimensions. For reference, Table A2 reports the bounding-box extents of the coarse surfaces for all *Shelly* scenes.

Hyperparameters and scheduling. Unless specified otherwise, we use learning rates similar to those commonly used in 3D Gaussian splatting [Kerbl et al. 2023]. Specifically, we set the zeroth-order SH, higher-order SH residual, and line-opacity learning rates to

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Table A1. Per-scene statistics for our *Fuzzy* dataset. All source images are captured at 6720×4480 px. COLMAP's image undistortion [Schonberger and Frahm 2016] crops images slightly differently per scene, and optimization is performed on a $4\times$ downsampled version of the undistorted images.

Scene	# Views	Orig. res.	COLMAP res.	Train res.
CACTUS1	173	6720×4480	6709×4466	1677×1116
CACTUS2	108	6720×4480	6712×4471	1678×1117
DINOSAUR	129	6720×4480	6711×4467	1677×1116
FLOWERS	122	6720×4480	6705×4459	1676×1114
FUR	128	6720×4480	6720×4474	1680×1118
KIWI	124	6720×4480	6715×4470	1678×1117
TAWASHI	124	6720×4480	6712×4470	1678×1117
TEXTILES	156	6720×4480	6707×4468	1676×1117

2.5×10^{-3} , 1.25×10^{-4} , and 5×10^{-2} , respectively. For positional optimization, we found that stable convergence is more reliably achieved using consistently small learning rates rather than aggressive annealing schedules. We therefore use initial positional learning rates that are $5\text{--}10\times$ smaller than those commonly used in 3DGS: 2.5×10^{-5} for the *Shelly* dataset and 1.6×10^{-5} for our *Fuzzy* dataset, exponentially decayed to $1/5$ of their initial values during optimization. The vertex-position learning rate is additionally scaled according to the camera extent, following standard 3DGS optimization practice. The weight of the length regularization $\mathcal{L}_{\text{len}}$ is set to 1.0×10^{-3} .

Details of baseline methods. For the 3DGS baseline [Kerbl et al. 2023], we sample the initial seed points using the same distance-band sampling around the same coarse proxy surface as our method for a fair comparison. While our method samples 1M seed points to initialize line primitives with 2M vertices, 3DGS directly uses 500K seed points to initialize 500K Gaussian primitives. For the DiffSoup baseline [Tojo et al. 2026], we initialize the triangle mesh using the MobileNeRF [Chen et al. 2023] mesh, reported in the original paper to produce the best results on the *Shelly* dataset. VolSurfs [Esposito et al. 2025] relies on implicit surface reconstruction and therefore does not require external seed points or primitive initialization.

Table A2. Axis-aligned bounding-box extents of the coarse proxy meshes for the *Shelly* dataset. We report the lengths of the shortest axis (Min.), longest axis (Max.), and the bounding-box diagonal (Diag.).

	KHADY	FERNVASE	KITTEN	PUG	WOOLLY	HORSE
Min.	1.14	1.24	0.92	2.23	2.99	0.80
Max.	1.67	1.93	2.24	2.55	3.62	3.57
Diag.	2.45	2.84	2.81	4.20	5.93	4.49

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